# Minxing Zheng

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**Research:** uncertainty quantification for machine learning, especially generative models and LLMs; trustworthy AI for high-stakes decisions.

**Seeking Ph.D. Research Scientist Internships, Summer 2027**

## Education

**Carnegie Mellon University**, *Ph.D. student* 2025–present

**University of Southern California**, *M.S., Statistics* 2022–2024

**University of Wisconsin–Madison**, *M.S., Statistics* 2020–2022

**Renmin University of China**, *B.S., Statistics; B.S., Applied Mathematics* 2017–2021

## Publications

*Conference and Journal Papers*

[C1] **Minxing Zheng** and Shixiang Zhu. Generative conformal prediction with optimized coverage allocation. In *Advances in Neural Information Processing Systems* (**NeurIPS**), 2026. [NeurIPS page] [project page].

[J1] **Minxing Zheng**, Xinran Miao, and Kris Sankaran. Interactive visualization and representation analysis applied to glacier segmentation. *ISPRS International Journal of Geo-Information*, 11(8):415, 2022. doi:10.3390/ijgi11080415.

*Preprints*

[P1] **Minxing Zheng**, Wenbin Zhou, and Shixiang Zhu. Beyond prediction: Conformal inference for latent distributional parameters. *arXiv preprint* arXiv:2608.03607, 2026. Short version accepted at the **NeurIPS 2026 Workshop on ML×OR**.

[P2] **Minxing Zheng** and Shixiang Zhu. Sample-efficient generative conformal prediction. *arXiv preprint* arXiv:2610.11349, 2026.

[P3] **Minxing Zheng**, Trevor Harris, and Shixiang Zhu. Decision-focused calibration with adaptive conformal predictive distributions. *Preprint*, 2026.

[P4] **Minxing Zheng**, Holly Wiberg, and Shixiang Zhu. Deciding when to decide: Testing operational suboptimality under distributional shift. *arXiv preprint* arXiv:2608.29465, 2026.

[P5] **Minxing Zheng**, Zewei Deng, Liyan Xie, and Shixiang Zhu. Learning to test: Latent representation learning for dynamic instability detection. *arXiv preprint* arXiv:2604.10967, 2026.

[P6] Xueqi Cheng, **Minxing Zheng**, Shixiang Zhu, and Yushun Dong. MISLEADER: Defending against model extraction with ensembles of distilled models. *arXiv preprint* arXiv:2506.02362, 2025.

[P7] Paromita Dubey and **Minxing Zheng**. Change point inference for non-Euclidean data sequences using distance profiles. *arXiv preprint* arXiv:2311.16025, 2023.

## Internship Experience

**Argonne National Laboratory**, Lemont, IL May–Jul. 2026

*Ph.D. Research Intern, ESI Grid Optimization and Analytics (Mentor: Feng Qiu)*

Developed conformal survival models to predict forced outages of hydropower units from equipment sensor telemetry and outage-event records, in collaboration with the Hydropower Research Institute (HRI). Manuscript in preparation.

## Honors and Fellowships

Heinz College Dean's Fellowship, Carnegie Mellon University 2025–2026

Marshall School Fellowship, University of Southern California 2022–2024

## Service

**Conference Reviewer:** NeurIPS, ICLR, ICML, AISTATS, AAAI

## Technical Skills

Uncertainty estimation and calibration for generative models · LLM fine-tuning and large-scale sampling · Differentiable optimization for decision-focused learning · Large-scale HPC experiments (Slurm, HTCondor) · Python (PyTorch, JAX), R, C++, MATLAB